# Fusion of Modalities and Evaluation of Multimodal Retrieval-Augmented Generation

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***Abstract*—Large Language Models (LLMs) such as ChatGPT and Gemini are used in many domains, but they mostly generate incorrect answers when detailed domain-specific questions are asked. This issue is called hallucination. Retrieval-Augmented Generation (RAG) reduces hallucination by retrieving relevant information from external sources and providing it as a context to the model. However, most existing RAG systems mainly works well with text data. In real-world applications, data is multimodal means it includes text, images, tables and charts. This makes multimodal retrieval and fusion challenging and poor fusion methods can reduce the accuracy of the system.**

**This research focuses on designing and evaluating multimodal retrieval-agmented generation (MRAG) pipelines using three fusion strategies: Early Fusion, Late Fusion and Hybrid Fusion. We implement three MRAG pipelines, each using one fusion technique and evaluate them on the same multimodal dataset. This study provides a comparative analysis of the three fusion-based MRAG pipelines and identifies which fusion strategy performs best under different evaluation conditions.**

***Index Terms*—Large Language Models (LLMs), Hallucination, Retrieval-Augmented Generation (RAG), Multimodal Retrieval-Augmented Generation (MRAG), Multimodal Data, Data Fusion, Early Fusion, Late Fusion, Hybrid Fusion, Multimodal Information Retrieval, Evaluation Metrics.**

## I. Introduction

Today, Large Language Models (LLMs) are used in every domain like healthcare, finance, retail, and so on. These models are trained on large-scale data available over the internet. However, large language models, such as ChatGPT, Gemini, Claude, etc., were not specifically trained to answer domain-specific questions.

It is observed that whenever LLMs are asked any in-depth domain-specific questions, they usually answer incorrectly or inaccurately with high confidence. This is known as hallucination[12]. To overcome this major limitation of Large Language Models, Retrieval-Augmented Generation (RAG) is used in practice[12].

RAG is used to provide additional context from a specific domain to the LLM from external sources like database, APIs, documents, file system, etc[12], [14], [15]. This context is further used to generate the response to user's questions. Retrieval-Augmented Generation performs well when data is expected to be textual, however in real-world problems data consists of different modalities including text, images, graphs, tables, audios, videos etc[6], [7].

Accuracy of RAG-based AI applications may degrade when data consists of different modalities. The degradation in performance depends on the methods used for fusing different modalities in the pipeline[6], [7]. Most RAG architectures are primarily designed for textual data, thus modifying the RAG pipeline to handle and combine multiple modalities usually may degrade the performance of RAG-based AI applications[4], [13].

The RAG pipeline is implemented in multiple phases and when it comes to Multimodal RAG, the most important phase is data fusion where we combine different modalities and provide context to the LLM for better response generation[6]. There are three data fusion techniques Early Fusion, Late Fusion, and Hybrid Fusion[6], [7].

In the context of Multimodal Retrieval-Augmented Generation (M-RAG), if the pipeline incorporates Early Fusion, vector embeddings are fused and stored in the same vector store. However, if the pipeline uses Late Fusion, vector embeddings from different modalities are stored and retrieved separately. Hybrid Fusion is a combination of Early and Late Fusion approaches.

This research paper studies different ways of implementing the Multimodal Retrieval-Augmented Generation pipeline along with different types of data fusion techniques and evaluation parameters and criteria for Multimodal Retrieval-Augmented Generation pipeline[6].

Finally, the research aims to implement three Multimodal Retrieval-Augmented Generation Pipelines, each incorporating different fusion techniques namely, Early Fusion, Late Fusion and Hybrid Fusion and evaluating all three pipelines on a single dataset on various parameters and criteria and concluding with comparative results.

## II. Related Work

In paper [1], Loo et al. present a Retrieval-Augmented Generation (RAG) system using GPT-4o-mini that supports flexible chunking, hybrid search and image retrieval. The authors explain that many RAG systems face two common problems. First, they use fixed chunk sizes, which may not work well for all types of documents. Second, they often ignore images inside documents, even though images can contain important information. To solve this, their system

allows users to choose the chunk size and overlap before converting the text into embeddings. They also extract images from PDFs and generate captions for them so that the system can retrieve images along with text. In their method, the PDF is divided into chunks, embeddings are stored in ChromaDB and relevant text chunks are retrieved using hybrid search. Then, the system compares the user query with the image captions to find related images. Finally, both the retrieved text and images are given to the LLM to generate the final answer. This paper shows that chunking plays an important role in improving retrieval accuracy and their approach follows a late-fusion style where text and image information are combined at the final stage.

In paper [2], Patel et al. propose a Multimodal RAG system that supports text, video and document-based question which give answers using the LLaMA3 model. The main issue which they address is that handling multiple types of data in a single RAG pipeline is difficult, especially when documents are non-parsable PDFs or contains handwritten content. Many existing MRAG systems fail in such cases because they cannot properly extract the required information. To solve this, the authors used OCR to extract text from scanned or handwritten PDFs. They also use a dynamic embedding alignment approach so that different modalities can be represented properly in the same retrieval process. In their pipeline, uploaded files are converted into base64 format and stored in Redis so that they can be accessed quickly, then extracted images are vectorized and stored in a Milvus vector database. When a user asks a query, the query is converted into embeddings, relevant content is retrieved from Milvus and the actual content is fetched from Redis along with chat history. Finally, all retrieved information is passed into a fine-tuned LLaMA3 model to generate the final response. Their system uses ImageBind embeddings to support multiple modalities and achieved around 92% retrieval accuracy which shows strong improvement compared to baseline RAG models.

In paper [3], a Multimodal Retrieval-Augmented Generation (MRAG) system designed for building context-aware chatbots. The authors noticed that extracting meaningful information from multimodal sources such as text, images and tables is still challenging, especially when users want specific information from large documents. They also mention that many chatbots don't handle context well and don't properly verify sources. Also, uploaded files usually work only in one chat, so users have to upload the same file again in other conversations. To solve these issues, the authors propose a system where uploaded files can be linked across multiple conversations, allowing the chatbot to reuse the same document context without repeated uploads. Their approach extracts text from documents, applies OCR to retrieve text from images and tables, converts the extracted content into embeddings and stores them in a vector database for retrieval. When a user asks a question, the query is also converted into embeddings, similarity search is performed and the retrieved context is passed to the LLM to generate the final response. The authors experimented with different models such as OpenAI models, LLaMA2 and Mistral and they finally chose Mistral because it offers better control. They also used EasyOCR to extract text and Milvus as the vector database because it can handle many requests at the same time which makes the chatbot system more scalable and fast.

Paper [4], is about a Structured Multimodal RAG framework to reduce hallucinations in LLM-based retrieval systems. Traditional RAG works well with text, but real-world data often includes images, tables and text, causing issues like semantic mismatches and irrelevant results. Existing MRAG systems handle images and text separately and focus mainly on retrieval, neglecting the generation stage which can produce hallucinations. To address this, Meng introduces three modules: Document Structure Enhancement to organize PDF layouts for better context understanding, Hybrid Retrieval and Reranking to combine sparse and dense retrieval with a cross-encoder reranker for improved relevance and Multimodal Feature Enhancement to integrate text and image information for high-quality responses.The framework is tested with metrics such as normalized Levenshtein distance, F1 and ROUGE-L, showing a structured method that reduces hallucinations while keeping the retrieval accurate.

In paper [5], Thiyagarajan presents a Multimodal RAG system for retail, aiming to improve information retrieval and generation for customer service and recommendation systems. The authors highlight that retail data comes from multiple sources and includes different types of data, such as text, images and structured information. Traditional systems often handle only one data at a time and cannot effectively combine these diverse data types. To address this, the paper proposes a cross-modal retrieval-augmented generation framework. Text, image and structured data are first converted into embeddings and processed with attention mechanisms. Then, cross-attention is applied between different modalities (e.g., text-image, image-structured, text-structured), followed by global fusion to combine all information. In the retrieval step it calculates a weighted similarity score across modalities to find the most relevant content, which is used by an LLM to generate responses. The system has three main components: (1) Data Integration, where text undergoes NLP processing and images are processed with CNNs for feature extraction, (2) Retrieval Component, which uses cross-modal attention to identify relevant knowledge base content and (3) Generation Component, where the final answer is produced by the LLM. This paper demonstrates that cross-modal attention and fusion can significantly improve retrieval and response quality in complex, multimodal retail datasets.

Paper [6], review multimodal data fusion techniques in smart healthcare. The authors point out that healthcare systems collect large amounts of diverse data, such as vital signs, wearable sensor readings, images and patient records, but combining these different types of data in proper format is a major challenge. To address this, they study three main fusion strategies. Early Fusion combines raw data from all modalities before training a single model, which is simple and uses full information but can lead to very large datasets and overfit-

ting. Late Fusion trains separate models for each modality and then combines their predictions; this approach is more flexible but may miss deep relationships between data types. Hybrid Fusion integrates data within a deep learning model, combining features in hidden layers, which allows learning complex relationships and achieves high performance but it is complex to implement and has a higher risk of overfitting. The paper compares these techniques using real healthcare studies, analyzes their advantages and disadvantages and discusses practical concerns such as data privacy and ethical considerations. Their review highlights that intermediate or hybrid fusion often provides the best performance, while late fusion remains effective for many practical systems.

In paper [7], Cheng et al. provide a analysis of multimodal data fusion. The authors point out that humans naturally combine multiple senses, like vision, hearing and touch, to understand the world, but AI systems often rely on only one type of data. Multimodal data fusion aims to combine information from different sources to improve AI understanding. The paper distinguishes two types of fusion: consistent information fusion, where multiple channels support the same conclusion and contradictory information fusion, where channels provide conflicting evidence. They introduce a confidence coefficient for each modality, representing how reliable that channel is, which allows the system to weigh different sources appropriately. The methodology includes mathematically modeling both consistent and contradictory cases, applying trainable confidence weights, and explaining the ideas using toy neural network examples. Although this paper is conceptual rather than proposing a full architecture, it provides practical insights into designing interpretable fusion strategies at both the feature and decision levels which makes it a useful reference for building multimodal systems that combine data reliably.

Paper [8], presents a study on multimodal data fusion using LLMs and attention mechanisms, applied to traffic systems. The traditional methods for combining multiple data types often suffer from low accuracy, slow processing and complex workflows. To address this, their approach first cleans and organizes raw traffic data from multiple sources, then extracts important features using an LLM to understand patterns and an attention mechanism to focus on the most relevant parts of the data. The attention mechanism helps the model ignore less useful information, preventing confusion from excessive data. The extracted features from different modalities are then combined to form a unified representation for downstream tasks. Their methodology also includes building mathematical and statistical models, applying recursive analysis and using a Gray model for prediction, followed by LLM-based feature extraction and weight optimization. Experiments show that this approach significantly improves performance compared to older methods, achieving around 87% accuracy and faster processing times. This paper demonstrates that combining LLMs with attention mechanisms provides an effective way to extract and fuse multimodal features.

In paper [9], Long et al. propose a Stepwise AI Interpretive Approach (SAII) for multimodal data fusion, Focusing on improving the interpretability of AI models that integrate text and images. The authors point out that traditional multimodal AI systems are often "black boxes" making it unclear which words or image regions influence the predictions, which reduces the trust in the model. Their approach analyzes image-text pairings, visualizes the model's focus using heatmaps, tests the importance of individual words and compares results before and after removing key words. Methodologically, the system uses the CLIP model to generate similarity scores between text and images, applies class activation mapping (CAM) to visualize attention, performs counterfactual testing by removing important words and evaluates impact on task performance. The study finds out that some background words or image areas can have strong influence on the model's results and their influence depends on neural activation thresholds. Although SAII is effective for small datasets, it requires further testing on larger, more complex systems. This work is important for MRAG systems as it provides a practical way to understand and interpret multimodal fusion, helping move AI from a "black box" to a "glass box" system.

This paper [10], proposes a novel Depth Mixed-ResNet (DM-RN) framework for knowledge graph-driven multimodal data fusion and analysis. The proposed method integrates Depth-First Search (DFS) for deep traversal of knowledge graph structures, ResNet for robust multimodal feature extraction and a Mixed Attention mechanism to enhance the model's ability to focus on salient spatial and channel-level information. By combining graph reasoning with deep residual learning and attention-based feature weighting, the DM-RN model effectively captures complex relationships among heterogeneous data sources. Experimental results demonstrate that the proposed approach outperforms conventional multimodal fusion models in terms of precision, recall, accuracy and F1-score, indicating improved representational capability and generalization performance in complex multimodal analysis tasks.

This paper [11], presents a new method to combine different types of medical data, such as MRI scans, PET scans, genetic information and clinical reports, to improve disease prediction. The proposed model combines Hypergraph Neural Networks, which help capture complex relationships between different data types, with an attention-based fusion technique that focuses on the most important information. The method was tested on the ADNI dataset for Alzheimer's disease and showed better performance compared to existing models in terms of accuracy and reliability. The results suggest that this combined approach can improve medical data analysis and support more accurate diagnosis and treatment planning.

The paper [12], study how reasoning abilities in Large Language Models (LLMs) impact Retrieval-Augmented Generation (RAG) systems. The LLMs often produce factually incorrect answers known as hallucination and that RAG can reduce hallucinations by retrieving relevant documents before answer generation. Their study focuses on whether LLMs with stronger reasoning skills can better utilize retrieved documents and produce more accurate responses. Methodologically, they

build a RAG pipeline using the PubMedQA medical question-answering dataset, indexing documents into embeddings stored in a vector database. When a user query is received, it is converted to a query embedding, the top-3 relevant documents are retrieved and the LLM generates an answer. They evaluate performance using Mean Reciprocal Rank (MRR) for retrieval quality, faithfulness to see if the LLM stays true to the retrieved content and Semantic Answer Similarity (SAS) to measure correctness at the meaning level. Three models are tested: o1-mini and o3-mini (reasoning-capable) and gpt-4o (non-reasoning). Their results show that models with better reasoning produce more accurate, faithful and semantically correct answers, highlighting the importance of reasoning capability in RAG pipelines.

In paper [13], Sun et al. propose a benchmark platform for evaluating Multimodal Retrieval-Augmented Generation (MRAG) systems. They point out that while LLMs like GPT have improved, they can still hallucinate and there is no standard way to measure the quality of RAG pipelines, especially for multimodal data (text and images). To solve this, they introduce MRAFE, which evaluates RAG systems in three steps: (1) Object Segmentation using YOLOv8 to split documents into headings, paragraphs, tables and images; (2) Context Synthesis, where text and images are converted into embeddings, combined and annotated with metadata; and (3) Vectorization and Storage for retrieval. The platform also generates domain-specific questions to test RAG systems and evaluates four key abilities: Noise Robustness (handling messy inputs), Negative Rejection (avoiding false information), Information Integration (combining retrieved and internal knowledge) and Counterfactual Robustness (keeping logical consistency in "what-if" scenarios). This work provides a systematic, standardized approach for benchmarking MRAG pipelines and helps developers compare and improve multimodal systems.

In paper [14], Kartiyanta et al. study how to make Retrieval-Augmented Generation (RAG) systems more cost-effective, especially for datasets like medical data. They explain that although large language models are powerful, they struggle with specialized topics and often rely on expensive embedding models for retrieval. To find a better solution, they compare different retrieval methods such as BM25 (sparse retrieval), dense retrieval with chunking and paid embeddings. They evaluate performance using metrics like recall, precision, MAP, BERTScore and RAGAS, and generate answers using GPT-4o-mini. Their results show that chunking is very important for dense models, BM25 works well for exact keyword matching and hybrid retrieval gives the best overall performance. They also conclude that paid embeddings are not always better and the best choice depends on the dataset and cost-performance balance.

In paper [15], Elkiran and Rasheed systematically evaluate how different similarity metrics affect the performance of Retrieval-Augmented Generation (RAG) systems. They explain that although RAG helps reduce hallucinations in large language models, its effectiveness depends heavily on how relevant documents are retrieved, which is controlled by the similarity metric. The authors compare three common metrics: Cosine similarity, Inner Product and L2 (Euclidean) distance. They build a complete RAG pipeline using the SQuAD dataset, store embeddings in a Milvus vector database with HNSW indexing, retrieve documents using the three metrics, rerank results with a cross-encoder model and generate answers using GPT-4o-mini. Performance is evaluated using retrieval metrics such as Recall, MRR, nDCG and Coverage, as well as completion metrics like latency, token usage and cost. Their results show that L2 distance performs poorly for RAG retrieval, while Cosine and Inner Product give much better results. They also find that retrieval quality greatly impacts overall RAG performance, while completion cost is mostly unaffected by the similarity metric. This study highlights that choosing the right similarity metric is very important for building efficient real-world RAG systems.

## III. Conclusion

We studied the problem of hallucination in Large Language Models and the limitations of traditional RAG systems when working with multimodal data[12]. While RAG helps improve accuracy by retrieving external information, most systems are mainly designed for text[4], [13]. In real-world applications, data includes text, images, tables and charts, which makes retrieval and fusion more difficult[6], [7].

To solve this, we built three Multimodal Retrieval-Augmented Generation (MRAG) pipelines using Early Fusion, Late Fusion and Hybrid Fusion techniques[6], [7]. All three pipelines were tested on the same dataset using the same evaluation methods to compare their performance.

The results show that the fusion method plays an important role in system performance[6]. Early Fusion combines all data at the beginning but may include noise. Late Fusion processes each modality separately and combines them at the end, which gives better control. Hybrid Fusion combines both approaches and provides a balanced performance in most cases.

Overall, this study shows that choosing the right fusion strategy is very important for building effective MRAG systems[6], [7]. In the future, this work can be extended by testing on larger datasets and improving fusion techniques to further increase accuracy and reliability.

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